

Project Executable Files

Date	03 October 2026
Team ID	4
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes — schema defined in app/database.py, created automatically on start; ER diagram in README
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	NA — local deployment as per project brief
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web application
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes — tests/ (245 pytest tests), Locust report and CPU/memory samples in reports/
10	Demo video or walkthrough (if required)	To be added by the team (link in repository README)

Step 2: File / Folder Structure

```
FitBuddy/
├── app/
│   ├── main.py           FastAPI app, static files, router, table creation
│   ├── config.py         .env settings (API key, models, admin password)
│   ├── routes.py         HTML + JSON routes
│   ├── database.py       SQLAlchemy models (users, plans) + CRUD helpers
│   ├── schemas.py        Pydantic models (UserInput, FeedbackRequest, WorkoutPlan)
│   ├── gemini_client.py  shared Gemini client, structured JSON, retry, fallback
│   ├── gemini_generator.py generate_workout_gemini()
│   ├── gemini_flash_generator.py generate_nutrition_tip_with_flash()
│   ├── updated_plan.py   update_workout_plan()
│   ├── demo_data.py      demo-mode plans, tips, revisions
│   ├── templates/        base, index, result, _plan, all_users, admin_login, error
│   └── static/           css/style.css, images/gym-bg.jpg
├── tests/               test_routes.py, test_security.py, test_ai_hardening.py,
                        test_demo_safety.py, conftest.py, locustfile.py,
├── resource_monitor.py
├── reports/             Locust report, perf summary, CPU/memory samples, repo
├── audit, logs
├── screenshots/         page screenshots
└── .env.example .gitignore .gitattributes pytest.ini requirements.txt README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Local: http://127.0.0.1:8000 (API docs: /docs)
Login Credentials (Demo)	Admin dashboard (/view-all-users): password = value of ADMIN_PASSWORD in your .env (default in .env.example: admin123). End users need no login.
Platform / Hosting Provider	Local machine with Uvicorn (Windows, macOS or Linux)
Repository Link	https://github.com/sharukeshalagiri/FitBuddy
Demo Video Link	To be added by the team (link in repository README)

Step 4: Run Instructions

1. Install Python 3.11+ and clone the repository:
`git clone https://github.com/sharukeshalagiri/FitBuddy.git`
2. Open the repository folder (the app is at the repository root):
`cd FitBuddy`
3. Create and activate a virtual environment:
`python -m venv venv`
`venv\Scripts\activate` (Mac/Linux: `source venv/bin/activate`)
4. Install dependencies:
`pip install -r requirements.txt`
5. (Optional, for real AI) copy .env.example to .env and set GEMINI_API_KEY (free key from <https://aistudio.google.com/apikey>). Without a key the app runs in demo mode.
6. Start the server:
`uvicorn app.main:app --reload`
7. Open <http://127.0.0.1:8000>, generate a plan, submit feedback, and visit </view-all-users> (admin) and </docs> (API).
8. Run tests: `pytest`

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Google's gemini-3.8-flash sometimes returns HTTP 503 (high demand), adding 13–20 s	Automatic retry and fallback to gemini-3.5-flash-lite; or set the light model as primary in .env
2	Free-tier Gemini API has rate limits	Demo mode for testing; load test run in demo mode
3	SQLite and a single admin password suit local / small use	SQLAlchemy allows switching to PostgreSQL; add per-user accounts in a future phase